

VerifAI: A Multimodal Deepfake Detection Framework

From Static Classification to Adaptive Multimodal Deepfake Detection

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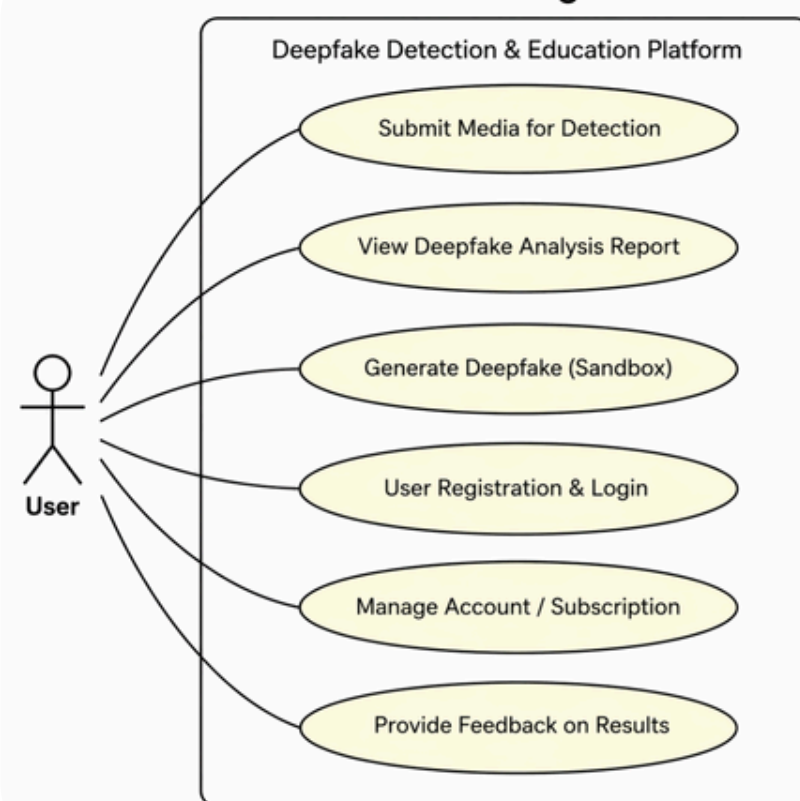
Introduction

Deepfakes erode digital trust, and current detection methods often face challenges with generalisation, handling missing modalities, and robustness in real-world scenarios. VerifAI overcomes these issues by employing adaptive multimodal detection that integrates feature-level fusion with dynamic inference routing to deliver a deployable, explainable system.

Objective

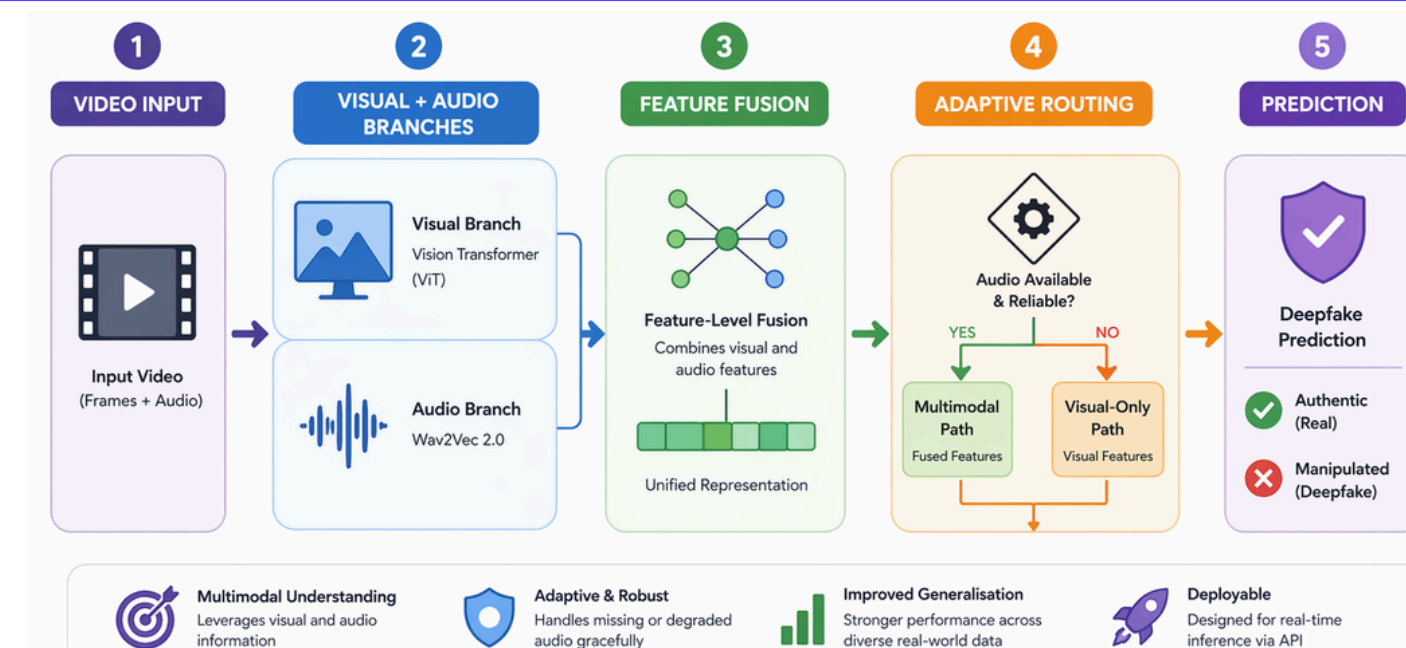
This project focused on developing and evaluating a multimodal deepfake detection framework that improves cross-dataset generalisation, incorporates adaptive inference routing, and ensures reliable real-world deployment through systematic testing and API integration.

UML Use Case Diagram



Methodology

- Problem Analysis and Literature Review:** Defined system requirements based on identified limitations in generalisation, multimodal detection, and explainability.
- Multi-Dataset Preprocessing:** Combined diverse datasets, extracted visual and audio features, and conducted subject-level data partitioning.
- Adaptive Multimodal Inference:** Combining feature-level fusion with dynamic routing to enhance detection robustness across different input scenarios.
- Experimental Evaluation:** Confirmed performance via benchmark tests, cross-dataset comparisons, and real-world experiments.
- Deployment and User Evaluation:** Deployed the system as an API prototype to evaluate usability, trust, and practical use.



Results

Experimental evaluation showed excellent detection accuracy, better generalisation across different datasets, and stable performance under various input conditions. The use of multimodal fusion and dynamic inference routing improved resilience to missing or poor-quality data, while steady convergence during training reflected effective optimisation and good generalisation.

Accuracy: 99%

ROC-AUC: 0.996

MCC: 0.84

Analysis

Grad-CAM analysis revealed that the model focuses on relevant manipulation cues rather than spurious artefacts, enhancing explainability and trust. User feedback showed high usability and confidence in the system's outputs. Additionally, the proposed adaptive multimodal framework, particularly the dynamic inference-routing component, played a crucial role in making deepfake detection more robust and practical.



Conclusion

VerifAI demonstrates that effective deepfake detection depends not only on model accuracy but also on an adaptive system design that incorporates multimodal reasoning, dynamic inference, and realistic testing environments. This approach improves robustness, user-friendliness, and deployment prospects, laying a foundation for future advancements in dependable deepfake detection.

Key Sources & Acknowledgements

Rössler et al., 2019 | Khalid et al., 2021 | Lyu, 2020 | Heidari et al., 2023 | Gong & Li, 2024

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